

Vinay Dinkar Kale

Portfolio: vinaykale64.github.io

Senior Data Scientist with 8+ years experience in Machine Learning with recent focus of contact center science domain.

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EXPERIENCE

- **Amazon (AWS)** *Santa Clara, USA*
Senior Data Scientist *July 2021 - Present*
 - Built novel Artificial Intelligence (AI) based forecasting and anomaly detection models for key contact center demand and supply metrics including contact volume, average handling time, and agent headcount for short-term, long-term, and intra-day applications for AWS Connect.
 - Developed Large Language Model (LLM) based AI agents to improve automation and customer experience in contact center manager and supervisor workflows.
 - Clients include several top Fortune 100 companies and over 300 government agencies who rely on our forecasting models to optimize agent allocation and reduce operational costs in their customer contact centers.
 - Mentored junior data scientists and drove adoption of latest state-of-the-art processes for continuous improvement.
- **Capital One** *NYC, USA*
Senior Data Scientist *Feb 2019 - July 2021*
 - Developed robust and reusable next-generation valuation, risk, and fraud XGBoost ML model pipelines reaching 100M+ customers, with an estimated incremental value of \$35M per year.
 - Architected an agile and automated monitoring framework for valuation models, decreasing the time required to complete quarterly compliance monitoring from one week to 8 hours.
 - Built Python libraries for data sourcing and cleaning with a focus on design, reusability, and user experience. Conducted multiple refactor cycles to maintain efficiency and keep systems up-to-date.
- **Spotify** *NYC, USA*
Data Science Intern *Jun 2018 - Aug 2018*
 - Developed a fan-artist pair segmentation pipeline that quantifies the affinity for 20 billion fan-artist pairs and analyzed how a new feature affects fan-artist journeys through statistical analysis on data from A/B tests.
- **ZS Associates** *Pune, India*
Data Scientist *June 2016 - July 2017*
 - Developed a marketing-mix modeling pipeline using multivariate regression for a client in the retail media sector. Increased total ROI by 15 percent and secured engagements for 3 additional similar projects.
 - Built a novel patient clustering engine for pharmaceutical clients using patient-data vector embeddings trained on sequential big medical data. This served as input to several patient-level classification models.

AWARDS AND RESEARCH

Patents

- US-12511576-B1: Correcting Time Series Predictions for Seasonality with Trends in Contact Centers (Granted)
- P86851-US01: Accurate Average Speed of Answer (ASA) and Service Level (SL) Forecasts using Effective Supply calculation in Contact Centers (Filed)

Publications

- Computer Vision and Pattern Recognition (CVPR) Conference 2018 : Traffic Surveillance Research Paper. We introduce novel usage of Mask-RCNN algorithm for vehicle detection, speed estimation and vehicle tracking for video footage of cars in highway. Achieved 100 percent detection rate and 7.97mi/hr speed estimation error
Citation Count: 15.
- Amazon Machine Learning Conference (AMLC) 2023: Generalized Anomaly Detection with Contact Center Use Case: A research paper reviewing various anomaly detection techniques and proposing a novel ensemble approach tailored for identifying anomalies in contact center environments.

Awards

- National Interest Waiver (NIW): Granted EB2-NIW by the US government to pursue the endeavor of "Improving efficiency of contact centers by using Artificial Intelligence (AI) driven Work-Force Management (WFM) systems" in June 2025.
- Runner Up, Columbia Data Science Society Hackathon: Built natural language processing network model to find fraud in Enron email data corpus. Won cash prize of 2,000 dollars. September 2017

EDUCATION

- **Columbia University** *NYC, New York*
M.S. Data Science *Sep 2017 - Dec 2018*
Relevant Coursework: Probability Theory, Statistics, Machine Learning, NLP, Deep Learning, Algorithms, Data Storytelling
- **Indian Institute of Technology Madras (IITM)** *Chennai, India*
B.Tech Mechanical Engineering, M.Tech Product Design *Aug 2011 - May 2016*
Relevant Coursework: Computational Engineering, Time Series Analysis, Regression Models, Operations Management

SKILLS

- **Languages** Python, R, SQL, Spark, C, C++, LaTeX
- **Frameworks** Scikit, Dask, TensorFlow, PyTorch, Tableau, MS Office, LLMs, Agentic AI.
- **Platforms** Docker, GIT, Airflow, MySQL, Linux, AWS, Microsoft Azure, Google Cloud

VOLUNTEER EXPERIENCE

- **Organizing Committee, PyData Conference NYC** *June 2019 - Sep 2019*
Led team for reviewing proposals and spearheaded the Diversity Scholarship Initiative
- **Teaching Assistant, Columbia University** *Jan-Apr/Aug-Dec 2018*
COMS 4995 Applied Machine Learning, COMS 4996 Applied Deep Learning.